People don't understand SpaceX

By Shaun Maguire · @shaunmmaguire · October 30, 2026

Mr. Pickles Lanman's TLDR:

Mixed

- Shaun Maguire, a Sequoia partner with a large SpaceX stake, publishes a model in which SpaceX revenue goes from about $51B in 2026 to about $731B in 2028.
- Nearly all of that growth comes from renting out AI computing capacity on the ground ("Terrestrial Compute"), not from rockets or Starlink.
- He argues SpaceX wins because of an unmatched team, a hard-to-copy Starship, and being the lowest-cost provider of AI compute.
- He says each Starship flight could support about $5B of lifetime Starlink revenue, and that phone-to-satellite service (Direct to Cell) will out-earn Starlink broadband by 2029 or 2030.

TRUTH CHECK The model's Q3 2026 starting figures match SpaceX's reported results, and its Q1 2027 forecast sits inside company guidance. Two supporting claims are off. SpaceX's 2015 milestone was a booster landing, not a catch. And today a Starship flight carries about 10 Falcon 9 launches' worth of Starlink capacity, not 20. His 2020 Starlink model, offered as proof of his track record, came in about 60% above actual revenue.

Serif text is the author's, unchanged. Underlines and numbers point to Pickles' margin notes.

Caveats: this is not investment advice. Sequoia Capital has a large position in SpaceX so I'm extremely biased. Also, Past performance is not indicative of future results.

After SpaceX's earnings call last week, I built a bottoms up financial model for the company. I haven't done this in over a year. The numbers that came out are breathtaking.

As the markets melt down today over fears of: a) political uncertainty with the midterms next week b) fears of an escalation in the Iran War c) some of the first massive agentic cyberattacks d) governments issuing prepping mandates e) sticky inflation that implies rates will likely go up in the short term, etc. It made me want to take the far other side and release my model, which will seem insane to 90%+ of investors. If nothing else it's fun to stir up a healthy debate :)

For those of you that don't know me, I've been privileged to be a major SpaceX shareholder since 2020 (having led Sequoia Capital's investments). I'm also a physicist, which is a huge advantage for underwriting SpaceX's next-gen projects. This is the Sequoia internal financial model that I built around Starlink in Q4 2020. Note that Starlink was in its first quarter of revenue when I built this. It seemed insane at the time.[1]

Year	2020	2021	2022	2023	2024	2025	2026
# of falcon 9 launches	19	20	24	28	36	44	52
# of starship launches	0	0	0	0	0	0	0
# of satellites launched	1102	1160	1392	1624	2088	2552	3016
Cumulative # of satellites (5 year lifetime)	1102	2262	3654	5278	7366	8816	10672
Download capacity per sat (in Gbps)	15	20	60	160	160	240	240
"oversubscription" multiple	1.0	2.0	2.0	3.0	3.0	4.0	4.0
Total effective capacity in Tbps	16.5	62.9	230.0	1009.5	2011.7	3403.7	5185.5
Useful capacity multiple	0.15	0.20	0.20	0.25	0.25	0.30	0.30
Total useful capacity in Tbps	2.5	12.6	46.0	252.4	502.9	1021.1	1555.6
Average "peak" usage in Mbps, per customer	10.0	12.0	14.4	17.3	20.7	24.9	29.9
Worst case # of users that can be supported (EC	247,950	1,048,833	3,194,028	14,604,890	24,254,075	41,036,482	52,098,303
ARPU per month	$60	$57	$54	$51	$49	$46	$44
EOY revenue run-rate estimate	**$178,524,000**	**$717,402,000**	**$2,075,479,250**	**$9,015,744,674**	**$14,223,668,909**	**$22,862,338,458**	**$27,573,868,981**
Annual revenue estimate (midpoint)	**$0**	**$447,963,000**	**$1,396,440,625**	**$5,545,611,962**	**$11,619,706,792**	**$18,543,003,684**	**$25,218,103,720**
Cost per satellite	$225,000	$202,500	$182,250	$164,025	$147,623	$132,860	$119,574
Cost per launch (launch only)	$28,500,000	$27,075,000	$25,721,250	$24,435,188	$23,213,428	$22,052,757	$20,950,119
Cost per launch (combined launch + satellites)	$42,000,000	$39,225,000	$36,656,250	$34,276,688	$32,070,778	$30,024,372	$28,124,572
Total annual launch cost (combined)	$798,000,000	$784,500,000	$879,750,000	$959,747,250	$1,154,548,013	$1,321,072,356	$1,462,477,764
# of Gateway sites	80	200	300	400	1400	2000	2500
Cost per gateway	$500,000	$500,000	$500,000	$500,000	$500,000	$500,000	$500,000
Total annual gateway cost	$40,000,000	$60,000,000	$50,000,000	$50,000,000	$500,000,000	$300,000,000	$250,000,000
Network costs (utilities, humans, etc)	$60,000,000	$66,000,000	$72,600,000	$79,860,000	$87,846,000	$96,630,600	$106,293,660
PP&E	$30,000,000	$55,000,000	$80,000,000	$250,000,000	$600,000,000	$800,000,000	$1,000,000,000
SG&A	$20,000,000	$100,000,000	$200,000,000	$400,000,000	$600,000,000	$800,000,000	$1,000,000,000
R&D	$100,000,000	$150,000,000	$200,000,000	$250,000,000	$300,000,000	$350,000,000	$400,000,000
Total costs before user terminals (UTs)	**$1,048,000,000**	**$1,215,500,000**	**$1,482,350,000**	**$1,989,607,250**	**$3,242,394,013**	**$3,667,702,956**	**$4,218,771,424**
EBITDA before UTs	-$1,048,000,000	-$767,537,000	-$85,909,375	$3,556,004,712	$8,377,312,779	$14,875,300,728	$20,999,332,296
# of user terminals shipped (# of new users)	247,950	800,883	2,145,194	11,410,862	9,649,185	16,782,407	11,061,821
Cost per user terminal	1,200	800	600	500	400	400	400
Total cost of user terminals	$297,540,000	$640,706,667	$1,287,116,667	$5,705,431,134	$3,859,673,997	$6,712,962,963	$4,424,728,331
EBITDA after UTs	-$1,345,540,000	-$1,408,243,667	-$1,373,026,042	-$2,149,426,422	$4,517,638,782	$8,162,337,765	$16,574,603,965

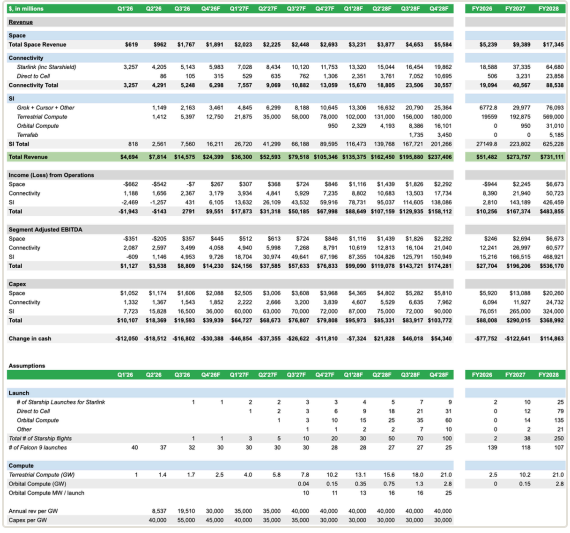
1 · TRUTH CHECK: MISLEADING

The 2020 model ran hot

The sheet he shows forecast Starlink revenue of $5.5B in 2023, $11.6B in 2024 and $18.5B in 2025. Actual segment revenue was $3.9B, $7.6B and $11.4B, about 60% below his forecast. He got the direction right, and his profit estimate came closer. But this is a weaker track record than the post suggests.

Midgard Finance, reported figures · Via Satellite on S-1

Here is my company wide forecast through 2028. There are wide error bars in certain areas, especially: revenue per gigawatt, CapEx per gigawatt and Terafab CapEx. But this is my 50th percentile forecast.[2]



$, in millions	Q1'26	Q2'26	Q3'26	Q4'26F	Q1'27F	Q2'27F	Q3'27F	Q4'27F	Q1'28F	Q2'28F	Q3'28F	Q4'28F	FY2026	FY2027	FY2028
Revenue															
Space															
Total Space Revenue	**$619**	**$962**	**$1,767**	**$1,891**	**$2,023**	**$2,225**	**$2,448**	**$2,693**	**$3,231**	**$3,877**	**$4,653**	**$5,584**	**$5,239**	**$9,389**	**$17,345**
Connectivity															
Starlink (Inc Starshield)	3,257	4,205	5,143	5,983	7,028	8,434	10,120	11,753	13,320	15,044	16,454	19,862	18,588	37,335	64,680
Direct to Cell		86	105	315	529	635	762	1,306	2,351	3,761	7,052	10,695	506	3,231	23,858
Connectivity Total	**3,257**	**4,291**	**5,248**	**6,298**	**7,557**	**9,069**	**10,882**	**13,059**	**15,670**	**18,805**	**23,506**	**30,557**	**19,094**	**40,567**	**88,538**
SI															
Grok + Cursor + Other		1,149	2,163	3,461	4,845	6,299	8,188	10,645	13,306	16,632	20,790	25,364	6772.8	29,977	76,093
Terrestrial Compute		1,412	5,397	12,750	21,875	35,000	58,000	78,000	102,000	131,000	156,000	180,000	19559	192,875	569,000
Orbital Compute								950	2,329	4,193	8,386	16,101	0	950	31,010
Terrafab											1,735	3,450	0	0	5,185
SI Total	818	2,561	7,560	16,211	26,720	41,299	66,188	89,595	116,473	139,768	167,721	201,266	27149.8	223,802	625,228
Total Revenue	**$4,694**	**$7,814**	**$14,575**	**$24,399**	**$36,300**	**$52,593**	**$79,518**	**$105,346**	**$135,375**	**$162,450**	**$195,880**	**$237,406**	**$51,482**	**$273,757**	**$731,111**
Income (Loss) from Operations															
Space	-$662	-$542	-$7	$267	$307	$368	$724	$846	$1,116	$1,439	$1,826	$2,292	-$944	$2,245	$6,673
Connectivity	1,188	1,606	2,367	3,179	3,934	4,841	5,929	7,235	8,802	10,683	13,503	17,734	8,390	21,940	50,723
SI	-2,469	-1,257	431	6,105	13,632	26,109	43,532	59,916	78,731	95,037	114,605	138,086	2,810	143,189	426,459
Total	**-$1,943**	**-$143**	**2791**	**$9,551**	**$17,873**	**$31,318**	**$50,185**	**$67,998**	**$88,649**	**$107,159**	**$129,935**	**$158,112**	**$10,256**	**$167,374**	**$483,855**
Segment Adjusted EBITDA															
Space	-$351	-$205	$357	$445	$512	$613	$724	$846	$1,116	$1,439	$1,826	$2,292	$246	$2,694	$6,673
Connectivity	2,087	2,597	3,499	4,058	4,940	5,998	7,268	8,791	10,619	12,813	16,104	21,040	12,241	28,997	80,577
SI	-609	1,146	4,953	9,726	18,704	30,974	49,641	67,196	87,355	104,826	125,791	150,949	15,216	166,515	468,921
Total	**$1,127**	**$3,538**	**$8,809**	**$14,230**	**$24,156**	**$37,585**	**$57,633**	**$76,833**	**$99,090**	**$119,078**	**$143,721**	**$174,281**	**$27,704**	**$196,206**	**$536,170**
Capex															
Space	$1,052	$1,174	$1,606	$2,088	$2,505	$3,006	$3,608	$3,968	$4,365	$4,802	$5,282	$5,810	$5,920	$13,088	$20,260
Connectivity	1,332	1,367	1,543	1,852	2,222	2,666	3,200	3,839	4,607	5,529	6,635	7,962	6,094	11,927	24,732
SI	7,723	15,828	16,500	36,000	60,000	63,000	70,000	72,000	87,000	75,000	72,000	90,000	76,051	265,000	324,000
Total	**$10,107**	**$18,369**	**$19,593**	**$39,939**	**$64,727**	**$68,673**	**$76,807**	**$79,808**	**$95,973**	**$85,331**	**$83,917**	**$103,772**	**$88,008**	**$290,015**	**$368,992**
Change in cash	**-$12,050**	**-$18,912**	**-$16,802**	**-$30,388**	**-$46,654**	**-$37,355**	**-$26,622**	**-$11,810**	**-$7,324**	**$21,828**	**$46,018**	**$94,340**	**-$77,752**	**-$122,641**	**$114,863**
Assumptions															
	Q1'26	**Q2'26**	**Q3'26**	**Q4'26F**	**Q1'27F**	**Q2'27F**	**Q3'27F**	**Q4'27F**	**Q1'28F**	**Q2'28F**	**Q3'28F**	**Q4'28F**	**FY2026**	**FY2027**	**FY2028**
Launch															
# of Starship Launches for Starlink			1	1	2	2	3	3	4	5	7	9	2	10	25
Direct to Cell					1	2	3	6	9	18	21	31	0	12	79
Orbital Compute						1	3	10	15	25	35	60	0	14	135
Other							1	1	2	2	7	10	0	2	21
Total # of Starship flights			1	1	3	5	10	20	30	50	70	100	2	38	250
# of Falcon 9 launches	40	37	32	30	30	30	30	28	28	27	27	25	139	118	107
Compute															
Terrestrial Compute (GW)	1	1.4	1.7	2.5	4.0	5.8	7.8	10.2	13.1	15.6	18.0	21.0	2.5	10.2	21.0
Orbital Compute (GW)							0.04	0.15	0.35	0.75	1.3	2.8	0	0.15	2.8
Orbital Compute MW / launch							10	11	13	16	16	25			
Annual rev per GW							8,537	19,510	30,000	35,000	35,000	40,000	40,000	40,000	40,000
Capex per GW							40,000	55,000	45,000	40,000	35,000	35,000	30,000	30,000	30,000

2 · TRUTH CHECK: CONFIRMED

Starting point matches reported results

Credit where due. His Q3 2026 figures match the reported ones: $14.6B revenue, $7.6B AI segment, $19.6B capex. His Q1 2027 revenue of $36.3B is inside the company's $33B to $37B guidance. His 10.2 GW of compute by end of 2027 is below Musk's stated hope of roughly 13 GW.

Q3 2026 call transcript

There are lots of nuances here. I'd be happy to debate any number.[3] Some of them can change dramatically with the macro, especially rev/GW. Note though that I believe rev/GW and CapEx/GW are coupled constants. Meaning if rev/GW goes down then CapEx/GW will also go down by a roughly proportionate amount. This would cause SpaceX's total GWs to go up as they're the low-cost provider, which would offset some of the revenue loss, but not by a fully proportionate amount (mainly power delays).

Barring a major disruption (war, AI moratorium, revolution) I think we are about to see a revenue ramp at SpaceX whose only parallel is NVIDIA's ramp[4] over the last three years. I believe that very few investors understand the company right now. In particular here are a few things that I think are under-appreciated.

3 · PICKLES' TAKE

The $200B cash hole goes unmentioned

His own sheet shows SpaceX burning about $78B of cash in 2026 and $123B in 2027 before turning positive in 2028. That is roughly $200B that has to come from borrowing or selling new shares. For shareholders, how that gets paid for matters a lot. The post never mentions it.

4 · PICKLES' TAKE

This is an AI cloud bet

In his model, ground-based compute rental brings in $569B of the $731B in 2028 revenue, about 78%. NVIDIA sells chips that its customers pay to deploy. SpaceX would be buying chips and building data centers itself, which takes far more capital. Readers should judge this as a bet on AI computing demand, not on rockets.

1. SpaceX has the best team of any company on Earth
1. Starship is the biggest moat in history
1. SpaceX is the low-cost provider in compute
1. The Starlink + Direct to Cell unit economics are astounding. And they improve dramatically with Starship

Let me unpack each of these.

Best team on Earth

China is cracked. I briefly worked at a start-up in Beijing in 2009, during the good old days before we became great power adversaries.



It is hard to explain the work ethic to Americans. The team when I started was 7 people and we all lived and worked in a 4 bedroom apartment on the outskirts of Beijing. It was 16 hour days 6 days

per week. It is a level of intensity that doesn't exist outside of China... except in Elon's companies.

SpaceX combines a max intensity culture with the best eval that exists in nature: does the rocket fly or explode? This last point is important and under-appreciated. It's probably only the AI community that will get the metaphor. It is impossible to survive at SpaceX if you're not up to the task. If a subsystem fails in a test all the way up to does the rocket explode, the root failure will be traced. Physics beats politics. SpaceX is the most extreme example of this.

When you then add in the importance of the mission and Elon's superpower for identifying and then empowering outlier young people... and then letting this stew and compound for 25 years against the backdrop of the ultimate eval... the result is by far the most elite team and culture on Earth.

What does this mean? Let's apply it to the AI buildout.

Amazon, Google and the Neoclouds have some great people. But *everyone* at SpaceX is great.[5]

5 · PICKLES' TAKE
Team claim rests on assertion
The team section is anecdote plus loyalty. No hiring, retention or productivity data is offered, and the claim cannot be tested. A strong culture is plausible. But none of it shows up as a specific input in the model, so it cannot support the revenue numbers.

Amazon, Google and the Neclouds have some people who can solve gnarly problems. But SpaceX (+ Tesla) has someone that can solve *any* problem.

Turbines are the bottleneck? SpaceX will make them.

Can't get a permit to turn on a power plant in Tennessee? No problem, turn on a power plant in Mississippi[6] and build high voltage transmission lines across the state border.

6 · PICKLES' TAKE
Left out: the permit fights
This refers to xAI's Southaven, Mississippi plant, which powers its Memphis data centers. xAI ran temporary gas turbines there for months without a permit. The NAACP has said it will sue under the Clean Air Act, and residents complain of pollution and noise. Counting this as an edge leaves out real legal and community risk.
CNBC

Having a network issue? Bring in the Starlink team.

Need to move a giant piece of equipment from Vietnam to Texas? Get the Starship recovery team on it.

We can play this game forever. People think of Gigawatt scale AI datacenters as the same level of difficulty as the tens of megawatt CPU datacenters of yesteryear. They're not. This is one to two orders of magnitude more challenging and it's why the SpaceX team is lapping everyone else.

The only other group of people that have experience dealing with atoms at this speed and scale are in China.

Starship is the biggest moat in history

What do you think are the most complex things humans have ever built? I've spent about a dozen hours walking around like a crazy person debating this with frontier models (voice mode!) The things the models normally come back with are: leading edge chip fab, ASML's EUV, ITER, nuclear submarines, the LHC and the ISS.

I think this is wrong, I think it's Starship. (Leading edge foundation models post RSI are getting close too)

Define it this way: if a group of the smartest humans with unlimited resources wanted to catch up to the frontier of the project, how long would it take?

Note that the answer to this depends on the supply chain you're starting with. If you're starting as cave men then leading edge fabs are almost certainly harder than Starship. But if you're starting with a reasonable assumption about the tools that would be available then I think it's usually Starship (with leading edge silicon being close and which is harder depends on debatable assumptions).

One way to evaluate this claim is to look at China. I'd argue they view semiconductors and space access as roughly equally strategically important. I'd further argue that they prioritized EVs, space, quantum and semis all around the same time.

In my opinion they surpassed the West in EVs in ~one decade. Quantum is much more nascent as a field and they caught up in a decade. For semis the rumors are that they've figured out early versions of EUV and are now perfecting them. I'd estimate China is about 5 years behind TSMC in semis. They are about 10 years behind SpaceX in launch (SpaceX did its first booster catch in 2015, China in 2026[7]). And I'd argue the gap is widening in space with Starship, while shrinking in semis.

This is splitting hairs though, the main point is that Starship is on the short list of machines at the frontier of human achievement. It's not something that can be replicated quickly. And SpaceX will continue to push the leading edge, which benefits America, unless the company is throttled (while China's space efforts will remain unthrottled).

> **7 · TRUTH CHECK: MISLEADING**
> **Landing, not catch, in 2015**
> SpaceX's first booster landing was December 21, 2015. Its first catch came years later. China first recovered an orbital booster on July 10, 2026, caught in a net at sea, then landed one on legs in August. So the roughly ten-year gap holds. China also brought two recovery methods online within weeks.
> Falcon 9 flight 20 · SatNews

Even if leading edge chip fabrication is harder, the Starship supply chain is vertically integrated within SpaceX. The Starship moat is just unfathomably large[8], evidenced by how far behind China is, even with the importance of space.

8 · PICKLES' TAKE
The moat still has to work
Starship's first try at catching both booster and ship is next month, and Musk calls success only "more likely than not." His model needs 38 Starship flights in 2027 and 250 in 2028, about five a week. Management targets weekly launches in early 2027. That gap is a big execution risk.
Q3 2026 call transcript

SpaceX is the low-cost producer in compute

Again, this is a point that we think is under-appreciated. Being the low cost producer in a commodity market is an extremely strategic position to sit in.

The price of a commodity can go below the lowest cost production cost for short periods of time, but not for very long. Quickly the high-cost producers go bankrupt, then a combination of: supply goes down and the low-cost producer consolidates more supply. Prices then go up and the low-cost producer actually comes away net stronger than before the pricing crunch.

Personally, I think the pricing dynamics of AI compute will be much closer to oil production rather than to broadband internet[9] (which was massively overbuilt in the dot com boom). The barrier to entry for Gigawatt scale AI compute is vastly higher, the supply chains are tighter, and the timelines are longer.

9 · PICKLES' TAKE
Oil keeps; chips lose value
The low-cost-producer logic holds for commodities. But oil in the ground holds its value, while AI chips lose value within a few years. If capacity is overbuilt, the losses come fast. That looks more like the broadband glut than oil. He also offers no cost comparison showing SpaceX really is the cheapest provider.

Even if the cost per gigawatt of compute went down to ~$10B I think SpaceX would come out ahead as the low-cost producer, and most agile player. I also think it's very unlikely we see anything below $20B/GW within the next ~4 years. And even at $20B/GW I think SpaceX's CapEx/GW goes down as the competition decreases and other producers go belly up.

Starlink/DTC/Starship Unit Economics

The current Starlink constellation is at about a $20B/yr of revenue run rate. This constellation was built with about 400 Falcon 9 launches. Each Starship flight will be the equivalent of about 20 Falcon 9 launches.[10]

10 · TRUTH CHECK: MISLEADING
Today it is about 10x
On the call, Musk said Flight 14's new V3 satellites delivered 26 Tbps of capacity, about 10 Falcon 9 launches' worth. The CFO described 20x as where capacity will ramp to. The roughly 400-launch figure is fair: about 436 Starlink launches total, with about 11,000 satellites working.
Q3 2026 call transcript · KeepTrack brief

That means it would take about 20 Starship flights to support $20B/yr in revenue. That's about $1B/yr of revenue per launch with about a 5 year lifetime of this revenue, or $5B in lifetime revenue per Starship flight.[11]

11 · PICKLES' TAKE
Capacity is not the same as revenue
This math assumes every bit of new capacity sells at today's prices. Revenue depends on customers, not satellites. His 2020 model made the same capacity-driven assumption and overshot. Costs are also missing: user dishes, ground stations and the satellites themselves. At today's 10x figure, the result roughly halves.

SpaceX will likely take costs down dramatically whether its for Starlink, Direct to Cell or Orbital Compute to increase the available market. But once you understand the scale of what each launch can provide something clicks.

Personally, I'd wager that Direct to Cell will do more revenue than Starlink in 2029 or 2030[12] under almost any scenario.

12 · PICKLES' TAKE
"Almost any scenario" is too strong
His own sheet shows Direct to Cell at about $105M last quarter, against $5.1B for Starlink. Even with his 2028 forecast of $23.9B, it would still trail Starlink's $64.7B. Overtaking within two more years takes everything going right, including phone and carrier deals. That is one scenario, not almost every scenario.

Conclusion

This is a very short post. I didn't justify most of my numbers. I'd be happy to do that in a live debate, this is the wrong format. And I'll acknowledge that there are wide error bars. But on the other hand, I feel very strongly that:

a) people are underestimating the revenue ramp that we are about to see

b) that people underestimate SpaceX's moat, and in particular how flexible Starship is

We didn't even talk about Orbital Compute as that really shows up starting in 2029. In a decade I believe that SpaceX will be as vertically integrated in the chip sector as it is today in the rocket sector. But we'll leave these topics for a future post.

The future won't be boring. We need to get it right. I believe that SpaceX will be critical infrastructure for any potential timeline.

*Again, this is not investment advice. And past performance is not indicative of future results.

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